Chapter 12. Identifiable Causal Discovery Models: LiNGAM

Graphical models, Lecture 5 (non-Gaussian errors, residual independence, ICA-LiNGAM, DirectLiNGAM, variants)

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1 Directions That Conditional Independence Cannot Distinguish

$$\boxed{X \to Y \to Z} \qquad \boxed{X \leftarrow Y \to Z} \qquad \boxed{X \leftarrow Y \leftarrow Z}$$

All three structures satisfy $X \perp\!\!\!\perp Z \mid Y$. Using conditional independence alone we find the skeleton $X - Y - Z$ but cannot tell the three directions apart (Ch. 9). Structural equations can add information:

$$X = BX + \epsilon, \qquad \epsilon_j \text{ independent.}$$

The *distribution* of the errors, not only their covariance, can carry direction information.

2 The LiNGAM Model

Motivation. With Gaussian noise, $X \to Y$ and $Y \to X$ fit the data equally well. If the noise is non-Gaussian the symmetry breaks: only in the true direction is the residual independent of the regressor, and this makes the whole causal order identifiable.

Definition: LiNGAM

Each variable is a linear combination of its parents plus a structural error:

$$X_j = \sum_{k \in \mathrm{pa}(j)} b_{jk} X_k + \epsilon_j, \qquad X = BX + \epsilon.$$

Assumption	Meaning
linearity	effects of the parents are linear combinations
acyclicity	with a suitable ordering of variables, B is strictly lower triangular
independent errors	$\epsilon_1, \ldots, \epsilon_p$ are mutually independent: no hidden common cause among the analyzed variables
non-Gaussianity	the structural errors are not normally distributed, which leaves information that distinguishes directions

Theorem: identifiability of LiNGAM (Shimizu et al., 2006)

Under linearity, acyclicity, mutually independent errors and non-Gaussian errors (at most one Gaussian error is allowed), the coefficient matrix B — hence the DAG and the causal order — is uniquely determined by the distribution of X.

3 In the True Direction, Cause and Regression Residual Are Independent

3.1 Two variables

True direction $X \to Y$:

$$X = \epsilon_X, \qquad Y = bX + \epsilon_Y, \qquad \epsilon_X \perp\!\!\!\perp \epsilon_Y, \quad \epsilon_X, \epsilon_Y \text{ non-Gaussian.}$$

Regressing Y on X leaves the residual $r_Y = Y - bX = \epsilon_Y$, so $X \perp\!\!\!\perp r_Y$.

Reverse direction $Y \to X$: fit $X = cY + r_X$. Then Y and r_X generally both contain ϵ_X and ϵ_Y, so for non-Gaussian data typically $Y \not\perp\!\!\!\perp r_X$.

Theorem: Darmois–Skitovich

Let $\epsilon_1, \dots, \epsilon_m$ be independent and $U = \sum_i \alpha_i \epsilon_i$, $V = \sum_i \beta_i \epsilon_i$. If $U \perp\!\!\!\perp V$, then every ϵ_i with $\alpha_i \beta_i \neq 0$ is Gaussian.

Theorem: the reverse regression leaves a dependent residual

In the model above with $b \neq 0$ and $\sigma_X^2 = \mathrm{Var}\epsilon_X$, $\sigma_Y^2 = \mathrm{Var}\epsilon_Y$, the least-squares coefficient is $c = \mathrm{Cov}(X, Y)/\mathrm{Var}Y = b\sigma_X^2/(b^2\sigma_X^2 + \sigma_Y^2)$ and

$$r_X = X - cY = \frac{\sigma_Y^2}{b^2\sigma_X^2 + \sigma_Y^2}\,\epsilon_X - c\,\epsilon_Y, \qquad Y = b\,\epsilon_X + \epsilon_Y.$$

r_X is uncorrelated with Y but, if either error is non-Gaussian, not independent of it.

Proof [Supplement]. $r_X = (1 - cb)\epsilon_X - c\epsilon_Y$ and $1 - cb = \sigma_Y^2/(b^2\sigma_X^2 + \sigma_Y^2)$. Both coefficients of r_X are nonzero, and both coefficients of Y are nonzero. If $r_X \perp\!\!\!\perp Y$, Darmois–Skitovich forces ϵ_X and ϵ_Y to be Gaussian. Uncorrelatedness is the normal equation of least squares. ■

If instead both errors are Gaussian, (r_X, Y) is jointly Gaussian and uncorrelated, hence independent: *both* directions produce independent residuals, which is exactly why Gaussian SEMs are not identifiable. **LiNGAM regresses in both directions and picks the direction in which explanatory variable and residual are more independent.**

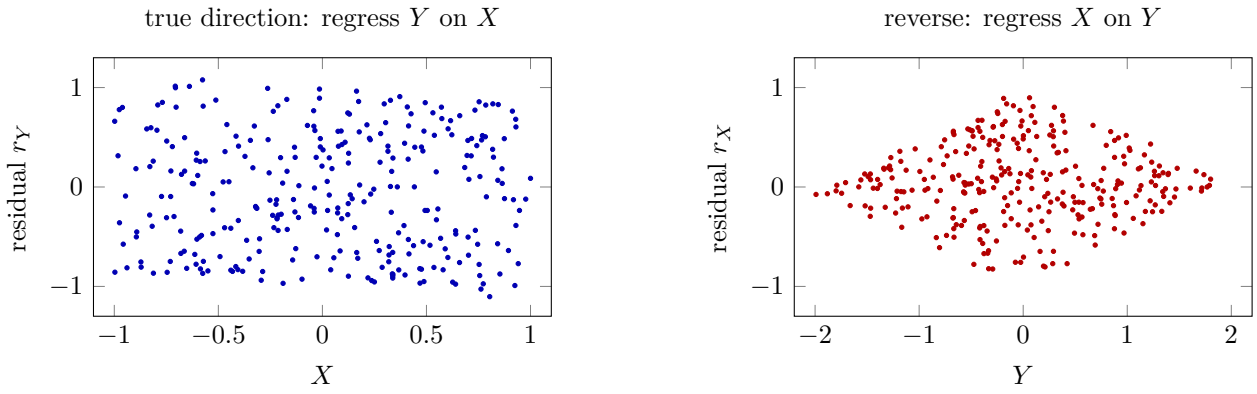


Figure 1: Simulation with $X = \epsilon_X$, $Y = X + \epsilon_Y$, $\epsilon_X, \epsilon_Y \overset{\text{iid}}{\sim} \mathrm{Unif}(-1, 1)$, $n = 300$. Both residuals are uncorrelated with their regressor (least squares guarantees it). In the true direction the cloud is a rectangle: $r_Y = \epsilon_Y$ is independent of X. In the reverse direction the cloud is a diamond whose vertical spread shrinks at the extremes of Y: uncorrelated but dependent (the correlation of the squares is 0.07 versus −0.42).

Example: a simulation picture

Take $\epsilon_X, \epsilon_Y \sim \mathrm{Unif}(-1, 1)$ and $Y = X + \epsilon_Y$. The scatterplot of (X, r_Y) is a rectangle (independent), while (Y, r_X) forms a rotated square (diamond) whose vertical spread of r_X shrinks at the extremes of Y: uncorrelated but clearly dependent.

4 Estimation

4.1 ICA-LiNGAM

Since B is permutable to strictly lower triangular form, $X = (I - B)^{-1}\epsilon = A\epsilon$: X is a linear mixture of independent non-Gaussian sources, the setting of independent component analysis (ICA).

Algorithm: ICA-LiNGAM

1. Run ICA to estimate an unmixing matrix W with WX having independent components; W equals $I - B$ up to an unknown row permutation and row scaling.
2. Find the row permutation of W giving no zeros (in practice, no small entries) on the diagonal; because $I - B$ has a unit diagonal and is triangular after ordering, this permutation is unique.
3. Divide each row by its diagonal element to get W', and set $\hat{B} = I - W'$.
4. Find the causal order by permuting $\hat{B}$ to be as close as possible to strictly lower triangular; prune small coefficients (e.g. by bootstrap or adaptive Lasso).

ICA is an iterative non-convex optimization and can converge to local optima; the result can depend on initial values.

4.2 DirectLiNGAM finds exogenous variables one by one

Theorem: characterization of an exogenous variable

In a LiNGAM, let $r_i^{(j)} = X_i - \dfrac{\mathrm{Cov}(X_i, X_j)}{\mathrm{Var}(X_j)} X_j$ for $i \neq j$. Then X_j is exogenous (has no parents) iff $X_j \perp\!\!\!\perp r_i^{(j)}$ for all $i \neq j$.

Proof ($\Rightarrow$) and idea ($\Leftarrow$) [Supplement]. If X_j is exogenous, $X_j = \epsilon_j$. Every $X_i = a_{ij}\epsilon_j + \sum_{l \neq j} a_{il}\epsilon_l$ with $A = (I - B)^{-1}$, and $\mathrm{Cov}(X_i, X_j)/\mathrm{Var}(X_j) = a_{ij}$, so $r_i^{(j)} = \sum_{l \neq j} a_{il}\epsilon_l$, independent of ϵ_j. Conversely, if X_j has a parent, then X_j contains the error of some ancestor, and some residual $r_i^{(j)}$ shares a non-Gaussian error with nonzero coefficients with X_j; Darmois–Skitovich then gives dependence (Shimizu et al., 2011). ■

Moreover, after regressing out an exogenous variable, the residuals $(r_i^{(j)})_{i \neq j}$ again follow a LiNGAM with the remaining causal order, so the procedure can be repeated.

Algorithm: DirectLiNGAM

For each candidate X_j, regress every other X_i on X_j to obtain residuals $r_i^{(j)}$, and add up the dependence between the candidate and all residuals:

$$T(j) = \sum_{i \neq j} \mathrm{Dep}(X_j, r_i^{(j)}).$$

1. Evaluate every variable as an exogenous candidate.
2. Choose the variable with the smallest $T(j)$ as the first in the order.
3. Replace the remaining variables by the residuals with the chosen variable's effect removed.
4. Repeat the same process on the remaining variables.

Dep is a measure of statistical dependence, e.g. a kernel-based independence measure or a mutual-information estimate. Unlike ICA, DirectLiNGAM has no iterative non-convex step and, given a consistent dependence measure, converges to the correct order in a fixed number of steps $(p - 1)$.

4.3 Repeating residual independence yields the causal order

Example: three variables

Let the data come from the LiNGAM

$$X_1 = \epsilon_1, \qquad X_2 = 0.8X_1 + \epsilon_2, \qquad X_3 = 0.7X_2 + \epsilon_3,$$

with $\epsilon_1, \epsilon_2, \epsilon_3$ mutually independent non-Gaussian errors.

Step	Independence observed	Choice
1	X_1 is independent of the residuals of X_2, X_3 after removing X_1	choose X_1 first
2	after removing X_1's effect, X_2 is independent of the residual of X_3	choose X_2
3	the last remaining variable	choose X_3

Result: $X_1 < X_2 < X_3$. (For candidate X_2 in step 1, the residual of X_1 on X_2 shares ϵ_1, ϵ_2 with X_2, so $T(2) > T(1)$.)

4.4 After the order: estimate direct edges and effects

Suppose DirectLiNGAM found $X_1 < X_2 < X_3$. Each variable uses only the variables before it as candidate parents:

$$X_2 = \beta_{21}X_1 + \epsilon_2, \qquad X_3 = \beta_{31}X_1 + \beta_{32}X_2 + \epsilon_3.$$

In the example $\hat{\beta}_{21} \approx 0.8$, $\hat{\beta}_{32} \approx 0.7$, $\hat{\beta}_{31} \approx 0$, so the estimated graph is

$$X_1 \Rightarrow X_2 \Rightarrow X_3.$$

Coefficients are estimated by least squares within the order, and small ones are pruned by sparse regression (adaptive Lasso) or tests; the order guarantees acyclicity.

5 Variants for Different Data Structures

Data or problem	Method	Caveat
independent cross-sectional data	DirectLiNGAM, ICA-LiNGAM	linear, acyclic structure, no hidden confounding
repeated measurements over time	VAR-LiNGAM	distinguish lagged and contemporaneous relations
feedback cycles	cyclic LiNGAM family	the acyclicity assumption of basic LiNGAM cannot be used
hidden common causes possible	ParceLiNGAM, RCD	estimate the identifiable part of the relations rather than the whole DAG

The LiNGAM variants are not applied one after another to patch a basic result; they are **chosen to match the data-generating structure**.

Remark: when LiNGAM is fragile

Near-Gaussian errors give weak signals, so large samples are needed. Measurement error, nonlinear effects, hidden confounders or selection bias break the independence of residuals and can make the wrong direction look better. Check the robustness of the order with

bootstrap resampling, and test the residual independence of the final model.

Summary

- Conditional independence leaves directions inside the Markov equivalence class undecided; non-Gaussian independent errors break the tie.
- In the true direction the cause is independent of the regression residual; in the reverse direction residual and regressor are uncorrelated but dependent (Darmois–Skitovich). With Gaussian errors both directions look the same.
- ICA-LiNGAM recovers B from the ICA unmixing matrix; DirectLiNGAM finds exogenous variables one by one through residual independence, then estimates coefficients within the order.

6 Selected exercises

Motivation. The exercises prove two steps used without proof by the estimation algorithms (the unique row permutation in ICA-LiNGAM and the recursion of DirectLiNGAM) and compute exactly the dependence visible in the simulation figure.

Exercise 12.1 (The row permutation in ICA-LiNGAM is unique)

Let $M = I - B$, where the variables admit a causal order κ (a bijection onto $\{1, \dots, p\}$) with $B_{jk} \neq 0 \Rightarrow \kappa(k) < \kappa(j)$.
(a) Show that the identity is the only permutation σ with $M_{\sigma(i),i} \neq 0$ for all i.
(b) Let $W = \Lambda P M$ with Λ diagonal invertible and P a permutation matrix (the ICA indeterminacy). Show that exactly one row permutation of W has no zero on the diagonal, and that it recovers M after dividing each row by its diagonal entry.

Solution [Supplement].

(a)

Step 1. $M_{jk} \neq 0$ requires $j = k$ or $B_{jk} \neq 0$, hence $\kappa(k) \leq \kappa(j)$.

Step 2. If $M_{\sigma(i),i} \neq 0$ for all i, then $\kappa(i) \leq \kappa(\sigma(i))$ for all i. The permutation $\tau = \kappa \circ \sigma \circ \kappa^{-1}$ of $\{1, \dots, p\}$ then satisfies $\tau(m) \geq m$ for all m. Since $\sum_m \tau(m) = \sum_m m$, the non-negative terms $\tau(m) - m$ sum to 0, so τ and hence σ is the identity.

(b)

Step 3. Row i of W is λ_i times row $\pi(i)$ of M, where π is the permutation encoded by P. Reordering the rows of W by a permutation q puts $\lambda_{q(i)} M_{\pi(q(i)),\cdot}$ in row i, whose diagonal entry is $\lambda_{q(i)} M_{\pi(q(i)),i}$.

Step 4. All diagonal entries are nonzero iff $\sigma = \pi \circ q$ satisfies $M_{\sigma(i),i} \neq 0$ for all i, i.e. by (a) iff $q = \pi^{-1}$. That reordering gives $\Lambda' M$ for a diagonal Λ', and since $M_{ii} = 1$, dividing each row by its diagonal entry returns M, hence $\hat{B} = I - M = B$. ■

Exercise 12.2 (The dependence in the simulation figure)

Let $X = \epsilon_X$ and $Y = X + \epsilon_Y$ with $\epsilon_X, \epsilon_Y \overset{\text{iid}}{\sim} \text{Unif}(-1, 1)$.
(a) Show that in the true direction $\text{Cov}(X^2, r_Y^2) = 0$.

(b) For the reverse regression $r_X = X - cY$, show that $c = \frac{1}{2}$, $\mathrm{Cov}(r_X, Y) = 0$ and $\mathrm{Corr}(r_X^2, Y^2) = -\frac{3}{7}$.

Solution [Supplement]. Write $u = \epsilon_X$, $v = \epsilon_Y$; then $\mathbb{E}u = \mathbb{E}u^3 = 0$, $\mathbb{E}u^2 = \frac{1}{3}$, $\mathbb{E}u^4 = \frac{1}{5}$, and likewise for v.

(a) $r_Y = Y - X = v$ is independent of $X = u$, so X^2 and r_Y^2 are independent and uncorrelated.

(b)

Step 1. $c = \mathrm{Cov}(X, Y)/\mathrm{Var}Y = \frac{1}{3}/\frac{2}{3} = \frac{1}{2}$, so $r_X = \frac{1}{2}(u - v)$ and $Y = u + v$. Then $\mathrm{Cov}(r_X, Y) = \frac{1}{2}(\mathbb{E}u^2 - \mathbb{E}v^2) = 0$.

Step 2. Since $(u - v)^2(u + v)^2 = (u^2 - v^2)^2$,

$$\mathbb{E}(r_X^2 Y^2) = \tfrac{1}{4}\mathbb{E}(u^4 - 2u^2v^2 + v^4) = \tfrac{1}{4}(\tfrac{2}{5} - \tfrac{2}{9}) = \tfrac{2}{45},$$
$$\mathbb{E}r_X^2\,\mathbb{E}Y^2 = \tfrac{1}{4} \cdot \tfrac{2}{3} \cdot \tfrac{2}{3} = \tfrac{1}{9}, \qquad \mathrm{Cov}(r_X^2, Y^2) = \tfrac{2}{45} - \tfrac{5}{45} = -\tfrac{1}{15}.$$

Step 3. Odd moments vanish, so $\mathbb{E}(u \pm v)^4 = \mathbb{E}u^4 + 6\mathbb{E}u^2\mathbb{E}v^2 + \mathbb{E}v^4 = \frac{16}{15}$ and $\mathrm{Var}\{(u \pm v)^2\} = \frac{16}{15} - \frac{4}{9} = \frac{28}{45}$. Hence $\mathrm{Var}(Y^2) = \frac{28}{45}$ and $\mathrm{Var}(r_X^2) = \frac{1}{16} \cdot \frac{28}{45} = \frac{7}{180}$, and

$$\mathrm{Corr}(r_X^2, Y^2) = \frac{-1/15}{\sqrt{(7/180)(28/45)}} = \frac{-1/15}{14/90} = -\frac{3}{7} \approx -0.43.$$

This matches the simulated value -0.42 in the figure: the residual of the reverse regression is uncorrelated with Y but its spread shrinks when $|Y|$ is large. ■

Exercise 12.3 (The DirectLiNGAM recursion)

In a LiNGAM $X = BX + \epsilon$ with $A = (I - B)^{-1}$, let X_j be exogenous and $r_i = X_i - a_{ij}X_j$ for $i \neq j$. Show that

$$r_i = \sum_{k \neq j} b_{ik} r_k + \epsilon_i \qquad (i \neq j),$$

i.e. $(r_i)_{i \neq j}$ is a LiNGAM with coefficient matrix B with row and column j deleted, the same errors $(\epsilon_i)_{i \neq j}$, and the induced causal order.

Solution [Supplement].

Step 1. $(I - B)A = I$ gives $A = I + BA$, i.e. $a_{ij} = \delta_{ij} + \sum_k b_{ik}a_{kj}$. Row j of B is zero (X_j is exogenous), so $X_j = \epsilon_j$ and $a_{jj} = 1$. Hence for $i \neq j$, $a_{ij} = b_{ij} + \sum_{k \neq j} b_{ik}a_{kj}$.

Step 2. Substitute $X_j = \epsilon_j$ and $X_k = r_k + a_{kj}\epsilon_j$ $(k \neq j)$ into the equation of X_i:

$$X_i = b_{ij}\epsilon_j + \sum_{k \neq j} b_{ik}(r_k + a_{kj}\epsilon_j) + \epsilon_i = \sum_{k \neq j} b_{ik} r_k + a_{ij}\epsilon_j + \epsilon_i.$$

Subtracting $a_{ij}X_j = a_{ij}\epsilon_j$ gives the claim.

Step 3. Deleting a row and the matching column keeps a matrix strictly lower triangular in the induced order, and $(\epsilon_i)_{i \neq j}$ are still mutually independent and non-Gaussian. So the assumptions of LiNGAM hold for the residuals, with the same direct effects b_{ik} among the remaining variables, and the characterization of exogenous variables can be applied again. ■